Machine Learning Questions and Answers

Q1. A bank wants to predict whether a loan application will be approved or rejected.

Answer: This is a Classification problem because the output is categorical (Approved or Rejected). Decision Tree is more suitable than KNN and Linear Regression because it can handle multiple applicant factors and provides clear decision rules.

Q2. Explain the difference between a feature and a label using a real-world example.

Answer: A feature is the input data used for prediction, while a label is the output we want to predict. Example: For house price prediction, house size, number of bedrooms, and location are features, while the house price is the label.

Q3. A Decision Tree predicts that a customer will purchase a product. Explain how the model reaches this prediction.

Answer: A Decision Tree asks a series of questions about the customer's characteristics, such as age, income, or purchase history. Based on the answers, it follows a path through the tree and reaches a final decision such as 'Will Purchase'.

Q4. Why is KNN considered a supervised learning algorithm? Can KNN perform classification if labels are not available?

Answer: KNN is a supervised learning algorithm because it learns from labeled data. It cannot perform classification without labels because it predicts a class based on the labels of the nearest neighbors.

Q5. Suppose a KNN model uses K = 5 and the nearest neighbors are: Yes, Yes, No, Yes, No. What will be the prediction and why?

Answer: The prediction will be 'Yes' because Yes appears 3 times while No appears 2 times. KNN chooses the majority class among the nearest neighbors.

Q6. Explain the equation of Linear Regression: $y = mx + b$. What do x, y, m, and b represent?

Answer: y is the predicted value (dependent variable), x is the input feature (independent variable), m is the slope of the line showing how much y changes when x changes, and b is the intercept, which is the value of y when $x = 0$.

Q7. What is correlation? What do the values +1, 0, and -1 indicate?

Answer: Correlation measures the strength and direction of the relationship between two variables. +1 indicates perfect positive correlation, 0 indicates no correlation, and -1 indicates perfect negative correlation.

Q8. A disease detection model produced the following result: Actual Condition = Disease Present, Prediction = Healthy. What type of prediction is this? Why can this be dangerous?

Answer: This is a False Negative prediction. It is dangerous because the patient actually has the disease but is predicted as healthy, which may delay treatment and worsen the condition.

Q9. What is the purpose of a Confusion Matrix? How is it different from a Correlation Matrix?

Answer: A Confusion Matrix evaluates the performance of a classification model by showing correct and incorrect predictions. A Correlation Matrix shows the relationships between variables using correlation values.

Q10. Compare the following algorithms in one or two lines each: Decision Tree, KNN, SVM, Linear Regression, Neural Network.

Decision Tree: Makes predictions using a tree of decision rules and is easy to interpret.

KNN: Classifies data based on the majority class of nearby data points.

SVM: Finds the best boundary that separates different classes.

Linear Regression: Predicts continuous values by fitting a straight line to data.

Neural Network: Uses interconnected neurons to learn complex patterns in data.